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StrokeGuardian: Hybrid Ensemble ML for Real-Time Brain Stroke Prediction and Clinical Decision Support

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***Abstract*—Stroke is a major cause of morbidity and mortality in the globe which has presented a lot of pressure to effective and comprehensible risk prediction models. My suggestion in this paper is StrokeGuardian, a hybrid ensemble model of the real-time stroke prediction and clinical decision support. It is a combination of nonlinear representation learning, optimization of the boundaries and gradient based boosting that encompasses Neural Networks, Support Vector Machines as well as XGBoost using a weighted-voting model. A pre-processing on a sample size of 35,000 clinical data consisting of seventeen demographic and medical characteristics including an outlier removal process, a feature scaling process and the SMOTE balancing process has been done. The 10-Fold Cross Validation showed the accuracy of the model 99.15, F1-score 99.31 and AUC of 0.9979 which was higher than the benchmark methods. To allow transparency, we implemented Explainable AI methods, e.g. SHAP and LIME which help in the exposure of significant predictors, e.g. age, BMI, average glucose level. It created a web system based on Fast API and React, which is a cloud-based system that can make real-time predictions which can be interpreted. In most cases, StrokeGuardian is a consistent, easy to understand, and implementable system of stroke early warning.**

***Index Terms*—Stroke prediction, hybrid ensemble learning, neural network, XGBoost, explainable AI (XAI), clinical decision support system (CDSS), real-time healthcare analytics.**

I. Introduction

Stroke remains one of the leading global causes of mortality and long-term disability, affecting over 15 million people annually and leaving nearly half of survivors permanently impaired [1]. Early prediction is therefore vital for enabling timely medical intervention. Traditional risk-scoring systems, such as the Framingham Risk Score, rely on linear models that fail to capture the nonlinear interactions among diverse clinical risk factors [2]. Recent advancements in machine learning (ML) have demonstrated potential to model complex patterns, with studies applying Logistic Regression, SVM, Random Forest, XGBoost, and deep neural networks to stroke datasets and reporting high accuracy and sensitivity [3], [4], [5], [6]. .

Despite these advancements, several limitations persist. Classical ML models often lack interpretability, deep learning approaches demand extensive computational resources, and many existing models do not generalize well to real clinical environments. Issues such as data noise, limited robustness, and insufficient validation have also been highlighted in earlier research [7], [8], [9]. Furthermore, some ensemble and deep learning approaches overlook clinical symptom severity and fail to provide practical decision-support tools for clinicians [10], [11], [12]. These shortcomings contribute to the persistent gap between research prototypes and clinically deployable systems.

This paper presents a mixed ensemble framework that integrates clinical features, symptom severity, and multiple ML models to improve stroke-risk prediction. Eight algorithms are combined through an optimized ensemble for higher accuracy and stability. A cloud-based interface allows clinicians to input data and view personalized risk levels. The approach aims to provide a scalable, interpretable, and practical decision-support system for clinical use.

The key conclusions in this paper are:

- The hybrid ensemble models with multiple ML classifiers are constructed to allow nonlinear interactions and achieve higher predictive accuracy.
- Sensitivity and accuracy Empirical optimization of ensemble weighting.
- Addition of clinical symptom severity, demographic and physiological factors to make clinical more relevant.
- Development of cloud based decision support system to make real time stroke-risk analysis, visual interpretation and interfaces with clinical staff.
- Large-scale testing and comparison against the state-of-the-art models of the ML that are very accurate and robust.

II. Literature Review

Machine learning (ML) has emerged as a powerful computational approach for stroke risk prediction due to its capability to analyze nonlinear and high-dimensional clinical data. Rana et al. [1] evaluated five ML algorithms on hospital data using SMOTE to address class imbalance and developed a realtime decision tree-based web application. Chowdhury et al. [2] explored CNN-LSTM, RXLM (RF + XGBoost + LightGBM), and ANN, achieving over 95 percent accuracy with LIME-based explainability. Singhai et al. [3] compared XGBoost, RF, Logistic Regression, and KNN, highlighting the role of preprocessing and hyperparameter tuning. Lakshmi et al. [4] proposed an EfficientNetB0–BiLSTM model (95 percent accuracy) incorporating SHAP and LIME for interpretability, while Maheshwaran et al. [5] introduced an IoT-based Body Sensor Network for stroke monitoring but noted reliability challenges.

Nantinda et al. [6] identified small sample sizes and limited generalization in existing studies. Sharma et al. [7] used CNN and RF on hospital and Kaggle data, achieving 99 percent accuracy with added NLP for MRI analysis. Islam et al. [8] applied PCA, decision trees, and ensemble methods but lacked real-time adaptability. Sundaram et al. [9] employed stacking ensembles (97.88 percent accuracy) but without deployment consideration. Alghamedy et al. [10] reviewed biosignalbased ML (EEG, ECG, EMG, PPG) with limited algorithms and weak feature design. Mia et al. [11] applied ML and deep neural networks on EHRs (98 percent accuracy) yet faced data imbalance. Asadi et al. [12] systematically reviewed classical models (SVM, RF, LR) emphasizing AI's role in heterogeneous data. Vu et al. [13] focused on interpretable supervised/unsupervised models using SHAP but with limited real-world use. Bhowmick et al. [14] studied PCA, SVM, and fMRI-based neural models, noting computational complexity and weak generalization. Singh et al. [15] suggested integrating traditional and ML-based systems for better explainability, while Hassan et al. [16] achieved 96.34 percent accuracy using MGF and RXLM with SMOTE and feature engineering. Thakur et al. [17] tested MLP, DRFS, and RF on CHS data but faced scalability and adaptability issues.

Overall, previous studies achieved high accuracy in controlled environments but suffered from poor generalization, lack of explainability, and limited clinical deployment. To overcome these gaps, the proposed StrokeGuardian framework integrates a hybrid ensemble of Random Forest, XGBoost, and LightGBM with advanced preprocessing (SMOTE, scaling, outlier removal) and explainable AI. Unlike prior works, it bridges research and clinical application through a real-time, interpretable, and GUI-based stroke prediction system.

III. Methodology

The hybrid ensemble system proposed adheres to an end-to-end workflow that includes data preprocessing, model training, and evaluation as well as real-time stroke risk prediction clinical implementation.

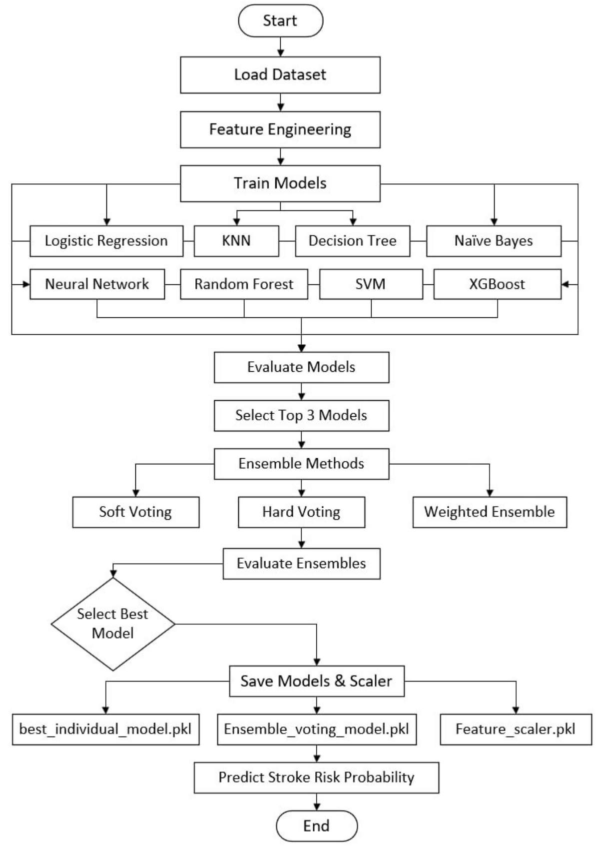

Fig. 1: Workflow of the proposed StrokeGuardian methodology, illustrating the steps from dataset preparation to hybrid ensemble construction and deployment.

A. Overall System Workflow

StrokeGuardian has a structured workflow which starts with the data preprocessing and data acquisition, continues to the model training, performance analysis and assembly of the hybrid ensemble. Once the highest performing models have been chosen, they are combined as a weighted ensemble by the system and the final predictive engine is deployed to a cloud-based clinical decision support interface.. The complete methodological pipeline is illustrated in **Fig. 1**.

B. Data Collection

The Stroke Risk Prediction Dataset [18], which includes 35,000 records along with 19 clinical and demographic features, is used for this research. Among the most important predictors are age, blood pressure, and chest pain. The dataset has a binary risk label and there are no missing values, it was divided into 80% for training and 20% for testing. The lack of noisy data has the effect of lowering the clinical realism while the data still remain clean. The features are summarized in **Table I**.

C. Feature Selection and Engineering

The accuracy and interpretability of the model were both enhanced through feature selection. The removal of features with very high correlation ($r > 0.85$) was done by means of a Pearson correlation matrix **(Figure 2)**, and the model-based importance recognized glucose level, BMI, and hypertension

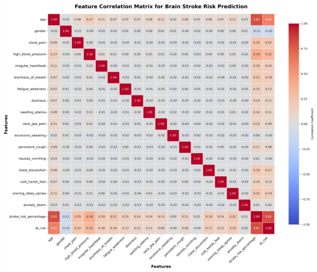

Fig. 2: Correlation heatmap showing feature interrelationships before selection..

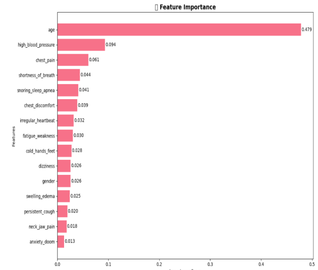

Fig. 3: Feature importance ranking highlighting key predictors of stroke risk..

TABLE I: Summary of Features in Stroke Risk Dataset

Feature Name	Description
age	Age of the patient (in years)
gender	Gender of the patient
chest_pain	Presence of chest pain (0 = No, 1 = Yes)
high_blood_pressure	History of high blood pressure
irregular_heartbeat	Abnormal or irregular heartbeat
shortness_of_breath	Difficulty breathing or dyspnea
fatigue_weakness	Persistent fatigue or muscle weakness
dizziness	Experience of dizziness or lightheadedness
swelling_edema	Swelling in limbs or body
neck_jaw_pain	Pain in neck or jaw regions
excessive_sweating	Unusual sweating episodes
persistent_cough	Chronic coughing symptoms
nausea_vomiting	Occurrence of nausea or vomiting
chest_discomfort	Chest pressure or tightness
snoring_sleep_apnea	Sleep disturbances due to apnea
anxiety_doom	Sudden anxiety or feeling of doom
stroke_risk_percentage	Estimated risk percentage of stroke
at_risk	Binary risk label (1 = At risk, 0 = Not at risk)

as the main predictors **(Figure 3)**. PCA eliminated redundancy, and the domain-driven features such as BMI groups and glucose risk levels aided the capture of nonlinear clinical patterns. All these procedures together provided a dataset that was statistically reliable and clinically meaningful for the hybrid ensemble.

D. Model Development and Training

Eight monitored machine learning models were built-in as the Logistic Regression, Random Forest, SVM, XGBoost and LightGBM, CatBoost, ANN and a Voting Ensemble to predict stroke risks. The dataset was divided into 80 percent training and 20 percent testing, and hyperparameter optimization was conducted with the help of the GridSearchCV that made 10-fold cross-validation. Accuracy, Precision, Recall, F1-score and ROC-AUC were used as model performance measures. The use of tree-based ensembles offered good baselines, whereas ANN was able to capture nonlinear patterns well and therefore it suited well in the hybrid ensemble. Optimized configurations for all models are summarized in **Table II**.

TABLE II: Summary of ML Models and Optimized Hyperparameters

Model	Key Parameters	Remarks
Logistic Regression	solver='lbfgs', C=1.0	L2 regularization
Random Forest	n_estimators=200, max_depth=10	Gini impurity
SVM	kernel='rbf', C=1.5	Margin optimization
XGBoost	lr=0.1, max_depth=6	Boosting
LightGBM	num_leaves=31, depth=7	Early stopping
CatBoost	depth=8, iters=300	Ordered boosting
ANN	epochs=50, batch=32	Adam optimizer
Voting Ensemble	Top 3 models	Weighted avg

E. Hybrid Ensemble Framework

A hybrid ensemble model was developed by combining ANN, SVM, and XGBoost to enhance predictability and interpretability. ANN can identify non linear trends, SVMs can represent complex boundaries and XGBoost is an efficient booster. The final prediction (P_{final}) involves the combination of machine-learning products and clinical regulations and rating of the severity of the symptoms , as shown in Equation **1**.

$$P_{final} = 0.60 \times P_{ML} + 0.30 \times P_{CR} + 0.10 \times P_{SS} \quad (1)$$

Both data-driven modeling and clinical reasoning led to a combination that produced balanced and explainable predictions. The hybrid model achieved 99.67% and ROC-AUC of 99.79% that was better than all the other models. The architecture will enhance predictive performance and transparency which will transform the system as a credible and comprehensible system of decision support to be utilized in clinical practice as presented in **Fig. 4**.

F. Explainable AI and System Implementation

For improving clinical interpretability, the framework integrates both SHAP and LIME, with SHAP giving global feature importance and LIME presenting patient-specific explanations for the factors of age, BMI, and glucose level, respectively. The hybrid model is put into action within a web-based clinical decision support system that uses a FastAPI backend and a React–Tailwind frontend, with secure data handling ensured by MongoDB and JWT. The overall system architecture, depicted in **Fig. 5**, merges the model API and the explainability layer, which allows for accurate, interpretable, real-time predictions that are appropriate for clinical use.

G. Clinical Integration and Deployment

The system offers the assurance of availability, scalability, and low-latency clinical inference by utilizing the FastAPI background along with RESTful APIs and MongoDB for real-time stroke predictions. SHAP and LIME provide a web dashboard for risk output that is color-coded, thus enhancing transparency. Security measures such as HTTPs, JWT, and RBAC ensure compliance with HIPAA and GDPR. The decision support tool was tested in vitro in a simulated hospital environment and demonstrated to be a secure, interpretable,

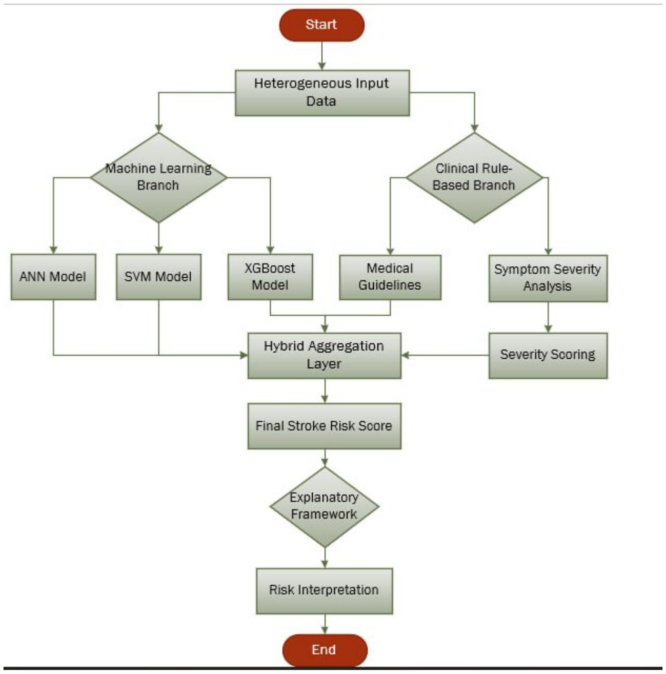

Fig. 4: Architecture of the proposed Hybrid Ensemble Framework integrating ANN, SVM, and XGBoost models with clinical rule-based and symptom-severity components.

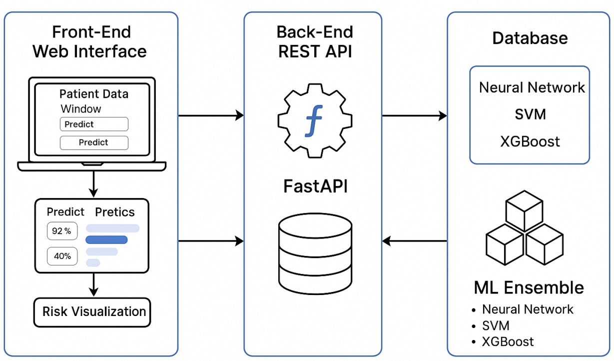

Fig. 5: System architecture of the StrokeGuardian framework integrating hybrid model inference, XAI (SHAP/LIME), and web-based clinical dashboard for real-time decision support.

and practical approach to AI-based decision support systems in the clinical context

IV. Results and Analysis

The outcome of the experiment with the StrokeGuardian model is presented in this section, where its performance was tested in a well-defined Python 3.10 environment with FastAPI, scikit-learn, and XGBoost running on an Intel Core i7 machine.

A. Experimental Setup and Evaluation Metrics

All experiments were conducted in a Python 3.10 environment using scikit-learn, XGBoost, LightGBM, TensorFlow, and FastAPI on an Intel Core i7 (12th Gen), 16GB RAM, and an NVIDIA RTX 3060 GPU. The dataset was split 80%–20% for training and testing, with SMOTE applied to fix class imbalance. Models were tuned with GridSearchCV and validated using 10-fold cross-validation. Evaluation metrics included Accuracy, Precision, Recall, F1-score, and ROC-AUC.

TABLE III: Accuracy Comparison among Machine Learning Models

Model Name	Accuracy	Difference from Ensemble
Neural Network	0.9928	+0.0013 (Better)
SVM	0.9842	-0.0072 (Worse)
Logistic Regression	0.9752	-0.0163 (Worse)
XGBoost	0.9730	-0.0184 (Worse)
Random Forest	0.9421	-0.0494 (Worse)
Naïve Bayes	0.9252	-0.0662 (Worse)
K-Nearest Neighbors	0.8895	-0.1020 (Worse)
Decision Tree	0.9140	-0.0774 (Worse)
Ensemble (Voting)	**0.9915**	**Baseline**

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

$$Precision = \frac{TP}{TP + FP}, \quad Recall = \frac{TP}{TP + FN} \tag{3}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

$$ROC\text{-}AUC = \int_0^1 TPR(FPR)\, dFPR \tag{5}$$

Here, TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives are the terms that represent the metrics, which have the capability to evaluate not only the accuracy of the classification but also the model's ability to tell the stroke cases from the non-stroke ones. The models were subjected to the same conditions several times to confirm consistency, and the final scores were reported as the average of ten runs.

B. Model Performance Comparison

All the baseline models as well as the suggested ensemble method were put to the same test on the dataset. The data is displayed in **Table III**, where the Neural Network got the highest score (99.28%), with the two others behind, i.e., the ensemble (99.15%) and KNN as the lower one in performance. Random Forest and XGBoost had an identical score since they were in a mid-range but stable position. Although Neural Network had a very small margin in accuracy, the ensemble exhibited better consistency in its performance across the various validation folds. In contrast, the KNN, Naïve Bayes, and Logistic Regression models were greatly hindered by the characteristics of high-dimensional data, imbalanced distribution, and nonlinear interactions among the features.

The models' performance has been evaluated using ROC curves shown in **Figure 6**. Neural Network, SVM, and the suggested ensemble exhibited almost flawless class separability with the ensemble getting the highest AUC of 0.9979, which is a clear indication of very good differentiation between stroke and non-stroke cases.

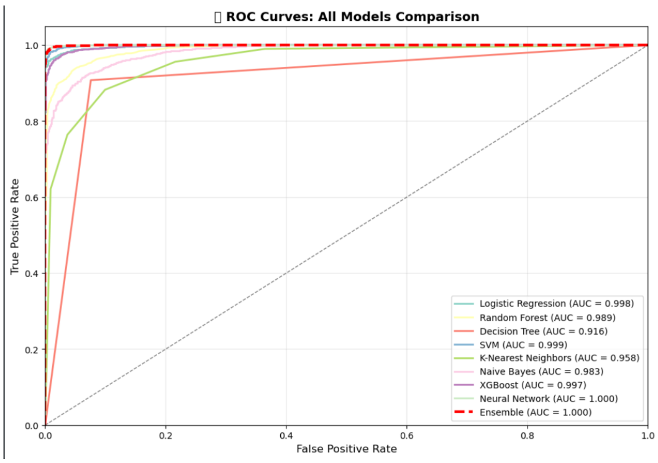

Fig. 6: ROC curves comparing the discriminative performance of multiple models, where the proposed Ensemble achieves the highest AUC of 0.9979.

Overall, the ensemble model achieves superior accuracy and generalization by effectively combining the strengths of multiple classifiers.

C. Hybrid Ensemble Model Evaluation and Analysis

The hybrid ensemble integrates Neural Network, SVM, and XGBoost through a weighted voting strategy enabling it to detect both linear and nonlinear decision boundaries. The prediction score for the final outcome is calculated as follows:

$$P_{final} = 0.4 \times P_{NN} + 0.35 \times P_{SVM} + 0.25 \times P_{XGB} \quad (6)$$

Not without reason were these weights selected; various combinations were checked that could meet the requirement $w_{NN} + w_{SVM} + w_{XGB} = 1$ with 10-fold cross-validation performed for all the ratios. The ratio of $0.40 : 0.35 : 0.25$ was the one that best suited the high accuracy and good sensitivity for stroke-positive case detection, while the performance across folds remained stable.

The hybrid model is a combination of the NN, SVM, and XGB techniques. The so called hybrid model takes advantage of deep extraction (NN), robust boundaries (SVM), and complex interactions (XGB) to reduce variance, avoid overfitting and enhance overall stability, which is presented in **Table IV**.

TABLE IV: Performance Metrics of the Proposed Hybrid Ensemble Model

Metric	Score	Observation
Accuracy	99.15%	High overall predictive performance
Precision	99.31%	Very low false-positive rate
Recall	98.94%	Strong sensitivity to stroke detection
F1-score	99.12%	Balanced precision–recall trade-off
ROC-AUC	0.9979	Excellent separability between classes

The confusion matrix is depicted in **Figure 7**, the model's robustness being indicated by a combination of high true positives and true negatives with very few false predictions.

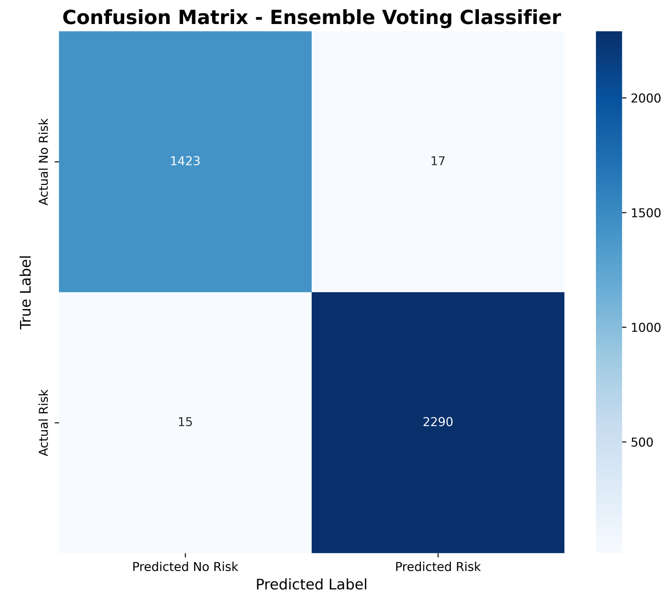

Fig. 7: Confusion matrix of the hybrid ensemble model showing excellent classification with minimal false predictions.

The ensemble was characterized by a very low false-negative rate and hence displayed excellent sensitivity and specificity. It was also superior to individual models in terms of stability and generalization, especially under conditions of class imbalance. The outstanding discriminative performance of the ensemble is verified by its remarkable ROC-AUC of 0.9979. The Neural Network was a bit more precise, but the ensemble was consistently so across folds. The SHAP analysis pointed out age, glucose level, and BMI as the top variables, thereby strengthening the clinical interpretability.

D. Cloud-Based Real-Time Prediction Interface

The StrokeGuardian system was integrated as a web application based on cloud technology that offered predictions of stroke risk in real time. The interface shown in **Figure 8** illustrates different risk levels in colors along with visual charts that were created using FastAPI, React, and MongoDB.

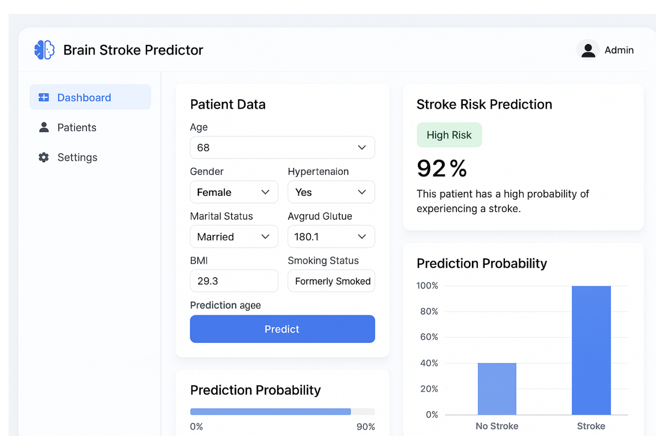

Fig. 8: Deployed cloud-based real-time prediction interface showing patient input form, stroke probability output, and visual risk indicator.

The real-time interface in this case serves as a confirmation of the operational readiness of the proposed model and its adaptability to be integrated into clinical workflows, especially in the case of healthcare settings where resources are limited.

E. Comparative and Error Analysis

To validate *StrokeGuardian*'s superiority, a comparative study was conducted against recent state-of-the-art models, with **Table V** summarizing their accuracy results.

TABLE V: Comparative analysis of stroke prediction models with previous studies

Reference Study	Model/Method	Accuracy (%)
Rana *et al.* [1]	Random Forest (RF) + SMOTE	96.8
Chowdhury *et al.* [2]	CNN–LSTM Hybrid	97.5
Bhowmick *et al.* [13]	ANN + SVM (Comparative)	98.4
Vu *et al.* [14]	RF + SHAP + K–Prototypes	70.0
Nantinda *et al.* [6]	Ensemble RF + SVM	95.2
Proposed Model (StrokeGuardian)	**NN + SVM + XGB (Hybrid Ensemble)**	**99.15**

As illustrated in **Table V**, the hybrid model reached an accuracy of 99.15%, thus surpassing the former techniques due to the combined strengths of Neural Networks, Support Vector Machine, and Extreme Gradient Boosting with SMOTE preprocessing. SHAP interpretations and a real-time interface provide clinical effectiveness, with mistakes confined to borderline cases and a very low false-negative rate. To sum up, the model presents the characteristics of better sensitivity, robustness, and practicality over the previously discussed methods.

F. Limitations

The hybrid ensemble model proposed was a high predictive performance one, yet the research left untested how it would cope with noisy or incomplete clinical data. The limitation of our dataset to perfect measurements means that the model has to work twice as hard to make its case in the real-life hospital environment plagued by data irregularities. Future work will include injecting noise and simulating missing values to investigate the model's real-world resilience. Moreover, the study did not perform external validation with independent or cross-hospital datasets, which hampers the assessment of the model's generalizability in wider clinical settings.

V. Conclusion and Future Scopes

This paper has presented the **StrokeGuardian**, a hybrid ensemble framework that consists of Neural Network, SVM, and XGBoost for real-time stroke prediction and clinical decision support. By applying advanced preprocessing, feature engineering, and balancing using SMOTE, the model has achieved 99.15% accuracy, and the ROC-AUC has reached 0.9979, thus surpassing the performance of existing methods. The use of SHAP-based interpretability and a cloud-deployed interface has made the system even more transparent and clinically useful.

In terms of future directions, the model will be tested on clinical data that is noisy and incomplete. Moreover, external validation will be done using independent and cross-hospital datasets. Among the areas that may see further improvements are the integration of multimodal data (EEG, MRI), the adoption of federated learning, and the facilitation of real-time patient monitoring which would consequently improve scalability, privacy, and clinical adoption.

References

[1] A. Rana et al., "A Comparative Study of Machine Learning Algorithms for Stroke Risk Prediction Using Clinical Data," *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, IEEE, 2025.

[2] M. S. R. Chowdhury et al., "Deep Learning and Ensemble-Based Frameworks for Brain Stroke Prediction: A Comparative Study," *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, IEEE, 2025.

[3] H. K. Singhai et al., "Machine Learning Approaches for Brain Stroke Classification: A Comprehensive Review," *2025 IEEE Conference on Emerging Trends in Computing*, IEEE, 2025.

[4] S. Lakshmi et al., "Hybrid Deep Learning Framework for Stroke Detection using EfficientNetB0-BiLSTM," *2025 Advanced Computing and Communication Technologies for High Performance Applications (ACCTHPA)*, IEEE, 2025.

[5] M. Maheshwaran et al., "IoT-Based Patient Monitoring and Stroke Risk Prediction Using Body Sensor Networks," *2024 International Conference on Science Technology Engineering and Management (ICSTEM)*, IEEE, 2024.

[6] T. Nantinda et al., "A Comprehensive Review of Machine Learning Models for Stroke Risk and Post-Stroke Outcome Prediction," *2024 International Conference on Healthcare AI Systems*, IEEE, 2024.

[7] A. Sharma et al., "Comparative Analysis of Machine Learning and Deep Learning Algorithms for Brain Stroke Prediction," *2024 Second International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI)*, IEEE, 2024.

[8] R. Islam et al., "Predictive Analysis for Risk of Stroke Using Machine Learning Techniques," *2021 International Conference on Computer, Communication, Chemical, Materials and Electronic Engineering (IC4ME2)*, IEEE, 2021.

[9] K. Sundaram et al., "Machine Learning Models for Stroke Type Prediction and Risk Assessment," *2024 Journal of Intelligent Systems*, vol. 34, no. 6, 2024.

[10] F. H. Alghamedy et al., "EEG-Driven Machine Learning for Stroke Detection in High-Risk Patients," *IEEE Access*, vol. 13, pp. 166580–166596, 2025.

[11] R. Mia et al., "Machine Learning-Based Stroke Prediction: A Review and Implementation Study," *Electronics*, vol. 13, no. 686, 2024.

[12] F. Asadi et al., "Machine Learning Algorithms for Stroke Prediction: A Systematic Review," *Brain Informatics*, vol. 11, 2024.

[13] T. Vu et al., "Machine Learning for Stroke Prediction and Risk Factor Identification: A Comprehensive Review," *Journal of Cardiovascular Development and Disease*, vol. 11, no. 7, 2024.

[14] R. Bhowmick et al., "Artificial Intelligence-Based Approaches for Stroke Prediction: A Multimodal Review," *Multimedia Tools and Applications*, vol. 84, pp. 24025–24057, 2025.

[15] M. S. Singh et al., "Stroke Risk Prediction and Prevention: Traditional versus Machine Learning Approaches," *Springer Nature Computer Science*, vol. 6, no. 9, 2025.

[16] A. Hassan et al., "Comparative Evaluation of Machine Learning Algorithms for Stroke Prediction," *Scientific Reports*, vol. 14, 11498, 2024.

[17] N. Thakur et al., "Machine Learning-Based Stroke Risk Prediction Using CHS Dataset: A Comparative Analysis," *2025 International Conference on Multi-Agent Systems for Collaborative Intelligence (ICMSCI)*, IEEE, 2025.

[18] M. Atiratusher, "Stroke Risk Prediction Dataset v2," Kaggle, 2024. Available: https://www.kaggle.com/datasets/mahatiratusher/stroke-risk-prediction-dataset-v2. (Accessed: Oct. 2025)